# MILITARY SPECIFICATION

## ADAPTER-BOOSTER, BOMB, TAIL, XM150 PARTS FOR

### 1. SCOPE

1.1 This specification covers the quality assurance provisions, and special requirements not covered by the drawings, for the fabrication of parts for one type of adapter booster designated as Adapter-Booster, Bomb, Tail, XM150.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.
MIL-C-45662 - Calibration of Standards.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.

FSC: 1325

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9240804 - Adapter-Booster, Bomb, Tail, XM150, Parts Arrangements

PUBLICATIONS

ET-9240804 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) 9240804 and with all requirements specified in applicable specifications.

3.3 Porosity of cast components.-Components fabricated from castings shall exhibit no visual evidence of porosity in excess of the following limits:

a. On non-threaded surfaces, internal and external, no individual pore shall exceed 1/4 inch in its longest dimension. The total of the maximum dimensions shall not exceed 3/4 inch. Pores 1/64 inch or less shall be disregarded.

b. No pore shall cause a discontinuity in the crest of a booster casing thread of the adapter exceeding 1/8 inch and shall not exceed 3/16 inch in other threads.

c. No pore shall extend entirely through a section.



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3.4 Workmanship.-All parts and subassemblies shall be fabricated and finished in a thorough workmanlike manner. The parts shall be free of burrs, sharp edges, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the



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certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to initiating regular production, the contractor shall submit ten (10) complete sets of parts (i.e., 10 each of each component) to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2.

The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures, and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-The components will be subjected by the Government to any or all of the examinations specified in 4.3.2 of this specification and any or all requirements of the applicable drawing.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon observing any nonconformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective



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action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeablity, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring



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activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD's listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Adapter (see dwg. 9240802 covering a detail of dwg. 9240804).

| Categories | Defects | Method of Inspection | Code No. (see 6.2) |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.00 percent | | |
| 101. | Pitch diameter of large external thread, maximum (max). | Gage | 01001 |
| 102. | Pitch diameter of internal thread, minimum (min)..... | Gage | 01002 |
| 103. | Concentricity of small external thread to internal thread..... | Gage | 01003 |
| 104. | Pitch diameter of small external thread, max..... | Gage | 01004 |
| 105. | Major diameter of large external thread..... | Gage | 01005 |
| 106. | Major diameter of small external thread..... | Gage | 01006 |
| 107. | Concentricity of inside diameter to thread..... | Gage | 01007 |
| 108. | Length of full internal thread, min..... | Gage | 01008 |
| 109. | Distance to drilled hole from indicated surface..... | Gage | 01009 |



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| | | | |
|---|---|---|---|
| Minor: | AQL 2.50 percent | | |
| 201. | Total length | Gage | 01010 |
| 202. | Wrench hole missing | Visual | 01011 |
| 203. | Protective coating missing or inadequate | Visual | 01012 |
| 204. | Excessive porosity (see 3.3), when applicable | Visual | 01013 |
| 205. | Marking missing, misleading or unidentifiable | Visual | 01014 |
| 206. | Evidence of poor workmanship (see 3.4) | Visual | 01015 |

4.3.2.2 Booster casing (see dwg. 9217222 covering a detail of dwg. 9240804).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Pitch diameter, min | Gage | 02001 |
| 102. | Minor diameter | Gage | 02002 |
| 103. | Length of full thread, min | Gage | 02003 |
| Minor: | AQL 1.50 percent | | |
| 201. | Total length, min | Gage | 02004 |
| 202. | Inside diameter, min | Gage | 02005 |
| 203. | Protective coating missing or inadequate | Visual | 02006 |
| 204. | Evidence of poor workmanship (see 3.4) | Visual | 02007 |



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4.3.2.3 Booster cup (see dwg. 9210420 covering a detail of dwg. 9240804).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Concentricity of large inside diameter with center tube..... | Gage | 03001 |
| 102. | Concentricity of counterbore in closed end with center tube.... | Gage | 03002 |
| 103. | Distance open end to counterbore bottom, min..... | Gage | 03003 |
| 104. | Distance through bottom at counterbore, max..... | Gage | 03004 |
| Minor: | AQL 1.00 percent | | |
| 201. | Outside diameter, max..... | Gage | 03005 |
| 202. | Evidence of poor workmanship (see 3.4)..... | Visual | 03006 |

4.3.2.4 Lead cup spacer (see dwg. 9217199 covering a detail of dwg. 9240804).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 1.00 percent | | |
| 101. | Pitch diameter of thread, max.. | Gage | 04001 |
| 102. | Diameter of lead cup hole..... | Gage | 04002 |
| 103. | Thickness, max..... | Gage | 04003 |
| 104. | Depth of counterbore..... | Gage | 04004 |
| 105. | Flatness..... | Gage | 04005 |
| 106. | Concentricity of lead cup hole with pitch diameter..... | Gage | 04006 |
| 107. | Thread damaged to extent assembly could be affected..... | Visual | 04007 |



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| | | |
|---|---|---|
| Minor: | AQL 1.50 percent | |
| 201. | Lead cup counterbore missing...Visual | 04008 |
| 202. | Wrench hole missing.....Visual | 04009 |
| 203. | Protective coating missing or inadequate.....Visual | 04010 |
| 204. | Evidence of poor workmanship (see 3.4).....Visual | 04011 |

4.3.3 Testing

4.3.3.1 Extruded booster cups (see dwg. 9210420) when booster cups are fabricated by the contractor "in-house" and Government acceptance is normally performed on site, the contractor has the option of assuring quality by tool control in lieu of sampling in accordance with Standard MIL-STD-105 and 4.3.2.3. Tool control shall be accomplished by the measurement of the first 5 consecutive pieces produced by the tooling with subsequent samples of 5 being taken as they come from the extrusion machine after approximately every 8 hours of production. All of the characteristics listed in 4.3.2.3 shall be examined by actual measurement of the characteristic under consideration. Go, no-go type gages shall not be used. Control charts shall be kept and maintained in the area of the extrusion process. Control limits for each characteristic shall be established which are acceptable to the Government. When the periodic examinations indicate that the limits are being exceeded or are likely to be exceeded in the next 8 hour period the contractor shall take necessary action to repair or adjust the tooling. A sample of 5 consecutive pieces shall be examined for acceptability of repair or adjustment before any production is permitted to continue.

With approval of the procuring agency, the contractor may institute this procedure at a sub-contractors plant if Government acceptance inspection is normally stationed at the sub-contractors plant.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-9240804 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.



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4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Calibration.-The contractor shall have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibrations of his acceptance inspection equipment. Records of such regularly scheduled calibrations shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments shall be in accordance with Specification MIL-C-45662.

4.4 Test methods and procedures.-There are no test methods or procedures required.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking

5.1.1 Level C.-Components shall be packaged and marked in accordance with Standard MIL-STD-1169. Components shall not be mixed in a container.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.



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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his



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operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5 digit Code Number contained in section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND11. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.



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6.4 Submission of First Article Samples.-Instructions as to the location for the evaluation of first article sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI, Dover, New Jersey 07801.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1325-A-341



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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